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Q1. Harshad number

ANSWER

```
n = int(input("Enter a number: "))
s = 0
t = n

while t > 0:
    s += t % 10
    t //= 10

if n % s == 0:
    print(str(n) + " is a Harshad number")
else:
    print(str(n) + " is not a Harshad number")
```

Q2. GCD

CODE

```
a=int(input("enter the number:"))
b=int(input("enter the number:"))
while b!=0:
    temp=b
    b=a%b
    a=temp
print("GCD is",a)
```

INPUT

```
10
20
```

OUTPUT

```
enter the number:enter the number:GCD is 10
```

Q3. sum to a target value

ANSWER

```
def find_pairs(lst, target):
    seen = set()
    pairs = []
    for num in lst:
        complement = target - num
        if complement in seen:
            pairs.append((complement, num))
        seen.add(num)
    return pairs

data = [int(x) for x in input("Enter numbers: ").split()]
t = int(input("Target: "))
result = find_pairs(data, t)
print("Pairs:", result)
```

Q4. Bitwise Operators

ANSWER

```
a=13, b=21, c=26, result=31
0b11111 0o37 0x1f
7 62 -32
#datatypes a,b,c,result:int
```

Q5. binary search

— Not Submitted —